WinInteropUtils

WinInteropUtils is a C# library written in .NET 8 that provides managed P/Invoke wrappers for tons of functions you can use, from reading file attributes to interacting with COM interfaces.

The base namespace is `FireBlade.WinInteropUtils`.

> **WARNING**
>
> WinInteropUtils is still in beta, so bugs may occur. If you experience any bugs, please report them on the Issues page.

Features

- WindowsFile file info wrapper
- Full enums (1600+ values) for HRESULTs and Win32 error codes
- Dialog boxes
- Useful utility User32 functions (SendMessage, SetWindowLongPtr...)
- Constants and Macros
- COM interfaces
- Icon management (DestroyIcon, ...)

Namespace FireBlade.WinInteropUtils

Classes

EnumExtensions
Provides extension methods for Enum.

ExceptionExtensions
Provides extension methods for Exception.

IconExtensions
Adds extension methods to the Icon class.

Macros
Defines Win32 macros.

MessageLoop
Provides methods for managing message loops.

Shell32
Declares P/Invoke functions from `Shell32.dll`.

StockIconHelper
Allows you to get a shell stock icon. Unlike the built-in .NET GetStockIcon(StockIconId, int) or GetStockIcon(StockIconId, StockIconOptions) functions, this wrapper allows you to enter any icon ID you want.

TitleBarInfo
Stores data about a window's title bar.

User32
Declares P/Invoke functions from `User32.dll`.

VisualStyle
Provides methods for managing and drawing with visual styles.

Win32Constants
Defines common Win32 constants.

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Window
Provides methods for managing Win32 windows.

WindowClass
Represents a window class.

WindowLongPtr
Defines default index values for SetWindowLongPtr(nint, int, nint) and GetWindowLongPtr(nint, int).

WindowStyles
Represents the various standard styles that can be applied to a window (after the window has been created, these styles cannot be modified, except as documented).

WindowsFile
Stores data about a Windows file.

ZOrderWindows
Represents windows used for Z position ordering.

Structs

CreateStruct
The structure passed to the `lParam` parameter of the `WM_CREATE` message of a window.

DrawTextAdvancedOptions
Represents a structure that contains extended formatting options

Shell32.SHFILEINFO
The internal Windows API struct for SHGetFileInfoW(string, uint, ref SHFILEINFO, uint, uint). Use with GetFileInfoEx(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO).

WinMessage
Provides information about a Windows message.

Interfaces

IHandle
Represents an interface for classes wrapping unmanaged handles.

Enums

ClassStyle
Represents class styles. Class styles define additional elements of the window class.

ExWindowStyles
Specifies extended window styles.

HRESULT

Defines common HRESULT values. Values starting with `S_` indicate success, while those starting with `E_` indicate an error.

MessageProcessingLimit
Specifies what message types should be processed by PeekMessage(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit).

MessageRemoval
Specifies message removal options for PeekMessage(ref WinMessage, Window?, uint, uint, MessageRemoval, MessageProcessingLimit).

Shell32.SHGetFileInfoFlags
Provides flags for the Windows API function SHGetFileInfoW(string, uint, ref SHFILEINFO, uint, uint). To use this function, call the GetFileInfoEx(string, WindowsFileAttributes, SHGetFileInfoFlags, ref SHFILEINFO) wrapper instead.

StockIcon
Icon identifiers for the GetIcon(StockIcon, StockIconOptions) function.

SystemColorId
Defines the identifiers for system colors.

SystemFontId
Defines the identifiers for system fonts.

ThemeTextOptions
Represents the options for DrawThemeText(Graphics, int, int, string, int, ThemeTextOptions, Rectangle).

TitleBarElement
Defines elements on a title bar.

TitleBarElementState
Defines the states of a title bar element.

Win32ErrorCode
Defines Win32 error codes.

WindowSetPosOptions
Window sizing and positioning flags for SetPos(Point, Size, WindowSetPosOptions, Window?).

WindowShowCommand
Represents the show command for Show(WindowShowCommand) and ShowAsync(WindowShowCommand).

WindowState

Defines states a window can be in.

WindowsFileAttributes

Specifies attributes for Windows files.

Delegates

WindowClass.WndProc

A callback function, which you define in your application, that processes messages sent to a window. The `WndProc` name is a placeholder for the name of the function that you define in your application.